

Improving Offshore Surface Gravity Wave Spectrum Predictions using Data Assimilation

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1 Theory

1.1 Linear Wave Theory

Ocean waves are a distinctive visual feature of the ocean surface, and very important to offshore operations such as wind farms. They are an instance of surface gravity waves. This section will outline the linear wave theory relevant to this project, as derived in ?, page 253.

Parameters describing surface gravity waves include the wavenumber k and direction θ , and the frequency ω . The first two describe the wave vector $\mathbf{k}$, which has magnitude k and direction θ . ω here denotes the absolute frequency measured in a fixed frame of reference. If the medium is subject to currents, this will differ from the relative frequency σ measured in a frame of reference moving with the mean current, due to the doppler effect. This difference is quantified by the equation

$$\omega = \sigma + \mathbf{k} \cdot \mathbf{U} \quad (1.1)$$

and σ and k are related by the dispersion relation

$$\sigma^2 = gk \tanh kD \quad (1.2)$$

The dispersion relation has two limiting forms:

- Deep water waves, where $kD \gg 1$, so $\tanh kD \approx 1$ and $\sigma^2 \approx gk$.
- Shallow water waves, where $kD \ll 1$, so $\tanh kD \approx kD$ and $\sigma^2 \approx gk^2D$

The phase speed is then given by

$$c = \frac{\sigma}{k} \approx \begin{cases} \sqrt{\frac{g}{k}} & \text{in deep water} \\ \sqrt{gD} & \text{in shallow water} \end{cases} \quad (1.3)$$

and the group speed may be expressed as

$$c_g = \frac{\partial \sigma}{\partial k} = \frac{1}{2}c \left(1 + \frac{2kD}{\sinh 2kD} \right) \approx \begin{cases} \frac{1}{2}c & \text{in deep water} \\ c & \text{in shallow water} \end{cases} \quad (1.4)$$

We can now use scale analysis to see that ω and σ generally have similar values. The ratio between $\mathbf{k} \cdot \mathbf{U}$ and σ in equation 1.1 is

$$\frac{\mathbf{k} \cdot \mathbf{U}}{\sigma} \sim \frac{kU}{\sigma} = \frac{U}{c}$$

Where $U = |\mathbf{U}|$ is the mean current speed. Since $c \gg U$ in general, this ratio is small and so $\omega \approx \sigma$.

Note that equations 1.1 - 1.4 are valid for linear waves of a single frequency and direction. Real ocean waves are more complicated than this, containing multiple frequencies propagating in multiple directions. The WAVEWATCH III model used in this project models ocean waves as a superposition of these simple linear waves.

Surface waves with multiple frequencies and directions are represented using a wave spectrum $F = F(\sigma, \theta)$, which is also a function of position and time. F represents the wave energy associated with a particular frequency σ and direction θ , so the total wave energy at any point in space and time is

$$E = \iint F d\sigma d\theta \quad (1.5)$$

Finally we define the wave action

$$N(\sigma, \theta) = \frac{F(\sigma, \theta)}{\sigma}$$

N is conserved in the absence of sources and sinks, making it the spectrum of choice for wave models.

Our observations come from acoustic doppler current profiler (ACDP) buoys, which estimate the spectrum by measuring the wave induced velocity and surface height variations. According to ?, the velocity and surface height anomaly induced by a simple wave with wave vector $\mathbf{k} = (k_x, k_y)$, frequency σ and amplitude a is

$$\begin{aligned} u &= a \frac{k_x}{\sigma} g \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \cosh k(z + D) \\ v &= a \frac{k_y}{\sigma} g \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \cosh k(z + D) \\ w &= -ia \frac{k}{\sigma} g \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \sinh k(z + D) \\ \eta &= a \exp i(\mathbf{k} \cdot \mathbf{x} - \sigma t) \end{aligned}$$

where only the real part of each expression should be taken. Note that the horizontal velocity (u, v) is always in the same direction as the wave vector $\mathbf{k}$. From the ADCP measurements, a timeseries of wave induced energy per unit area $E = \int_{-D}^{\eta} \frac{1}{2} \rho_0 \|\mathbf{u}\|^2 dz + \frac{1}{2} \rho_0 \eta^2$ can be calculated. This is equivalent/a good approximation to the wave energy in equation 1.5 (source) so a Fourier transform of E gives the wave spectrum integrated across all directions $E(\sigma) = \int F(\sigma, \theta) d\theta$ (how is directional distribution estimated?)

(Include ADCP locations if possible)

1.2 Model Description

Waves are modelled using WAVEWATCH III, which uses as its base spectrum a wavenumber-direction spectrum $F(k, \theta)$, rather than the frequency-direction spectrum described above. The spectrum is discretised using 29 sampled wavenumbers k and 24 angles θ . The wavenumber-direction spectrum is related to the frequency-direction spectrum by the Jacobian transform

$$F(\sigma, \theta) = \frac{\partial k}{\partial \sigma} F(k, \theta) = \frac{1}{c_g} F(k, \theta)$$

Wave propagation is then described by the equation

$$\frac{D}{Dt} N(k, \theta) = \frac{S}{\sigma} \quad (1.6)$$

Where $N(k, \theta) = F(k, \theta)/\sigma$ as before and S describes the net effect of sources and sinks for the spectrum $F(k, \theta)$.

In deep water, S consists of three main parts: an atmospheric interaction term S_{in} , a nonlinear wave-wave interaction term S_{nl} and a dissipation term S_{ds} .

S_{in} is dominated by an exponential wind-wave growth term, so S_{in} is equal to some coefficient multiplied by $\sigma N(k, \theta)$. This leads to a feedback which is usually positive but can be negative in the case of swell, and represents the main source of wave energy.

S_{ds} is generally dominated by wave breaking, and it represents a sink of wave energy. Similar to S_{in} , it is equal to a negative coefficient multiplied by $\sigma N(k, \theta)$. The balance between S_{in} and S_{ds} determines the overall growth or decay of wave energy.

S_{nl} represents interactions between waves due to nonlinear processes, which results in energy transfer between waves of different frequencies. (maybe expand - ?)

In shallow water, processes such as wave-bottom interactions, wave breaking and reflection off shorelines become important, so additional source terms are included to describe this.

As forcing, WAVEWATCH III uses just the 10 metre wind, and this comes from the GEFS ensemble. The GEFS forecast contains one control run plus 20 perturbed ensemble members. Each GEFS run acts as forcing for one member of the wave ensemble. (Do they?) The model is run on a rectangular grid at 0.5° globally. There are then two higher resolution nested grids covering the region of interest in the gulf of Mexico: a 0.1° resolution grid covering the whole gulf, and a 0.05° grid covering the specific region of interest. Since the main sources of dangerous waves in the region are hurricanes from the East and cold fronts from the North, the first nested grid is positioned so that the smaller nested grid lies in its southwest region. (Include figure - ask Irina)

1.3 Data Assimilation

As well as the model, we also have some direct observations of the waves which we would like to make use of to improve our predictions. We use techniques of data assimilation to achieve this.

We take a vector $\mathbf{x}$ to represent the state of the system, assuming it has a normal background distribution $\mathbf{x} \sim N(\mathbf{x}_b, \mathbf{B})$. We take a vector $\mathbf{y}$ to represent a set of observations, and assume $\mathbf{y}$ is related to $\mathbf{x}$ via an observation operator $\mathcal{H}$:

$$\mathbf{y} = \mathcal{H}(\mathbf{x}) + \varepsilon$$

where the error term ε is assumed to be normally distributed with mean zero and covariance $\mathbf{R}$: $\varepsilon \sim N(0, \mathbf{R})$. In our case $\mathcal{H}$ is linear i.e. there is a matrix $\mathbf{H}$ such that $\mathcal{H}(\mathbf{x}) = \mathbf{H}\mathbf{x}$, and so $\mathbb{P}(\mathbf{y}|\mathbf{x}) = N(\mathbf{H}\mathbf{x}, \mathbf{R})$. We can then express the distribution of $\mathbf{x}$ given some set of observations $\mathbf{y}$ using Bayes' theorem:

$$\mathbb{P}(\mathbf{x}|\mathbf{y}) = \mathbb{P}(\mathbf{y}|\mathbf{x}) \frac{\mathbb{P}(\mathbf{x})}{\mathbb{P}(\mathbf{y})} \quad (1.7)$$

This allows $\mathbb{P}(\mathbf{x}|\mathbf{y})$ to be determined since $\mathbb{P}(\mathbf{y}|\mathbf{x})$ and $\mathbb{P}(\mathbf{x})$ are known, and $\mathbb{P}(\mathbf{y})$ is independent of $\mathbf{x}$ so it acts as a normalisation constant. Since all the distributions involved are normal, $\mathbb{P}(\mathbf{x}|\mathbf{y})$ will be normally distributed also:

$$\mathbb{P}(\mathbf{x}|\mathbf{y}) \sim e^{-\frac{1}{2}(\|\mathbf{x}-\mathbf{x}_b\|_{\mathbf{B}^{-1}}^2 + \|\mathbf{y}-\mathbf{H}\mathbf{x}\|_{\mathbf{R}^{-1}}^2)} \quad (1.8)$$

adopting the notation $\|\mathbf{a}\|_{\mathbf{M}}^2 = \mathbf{a}^\top \mathbf{M} \mathbf{a}$. Finding the most likely value of $\mathbf{x}$ given this distribution is equivalent to minimising a cost function

$$J(\mathbf{x}) = \frac{1}{2} \|\mathbf{x} - \mathbf{x}_b\|_{\mathbf{B}^{-1}}^2 + \frac{1}{2} \|\mathbf{y} - \mathbf{H}\mathbf{x}\|_{\mathbf{R}^{-1}}^2 \quad (1.9)$$

The state that minimises this cost function is called the analysis $\mathbf{x}_a$ and may be expressed as

$$\mathbf{x}_a = \mathbf{x}_b + \mathbf{K}\mathbf{d} \quad (1.10)$$

where $\mathbf{d} = \mathbf{y} - \mathbf{H}\mathbf{x}$ is the innovation and $\mathbf{K} = \mathbf{B}\mathbf{H}^\top (\mathbf{H}\mathbf{B}\mathbf{H}^\top + \mathbf{R})^{-1}$ is the Kalman gain.

In our case, we have observations from multiple times $t = 0, 1, \dots, T$ spread out over some time window, and we expect the state $\mathbf{x}$ to evolve over this time window. This evolution may be predicted using the model as follows: let $\mathbf{x}_0$ represent the state of the system at $t = 0$, then we define the model operator $\mathcal{M}_t$ for each t such that $\mathbf{x}_{t+1} = \mathcal{M}_t(\mathbf{x}_t)$ as predicted by the model. The cost function then becomes

$$J(\mathbf{x}_0) = \frac{1}{2} \|\mathbf{x}_0 - \mathbf{x}_b\|_{\mathbf{B}_0^{-1}}^2 + \frac{1}{2} \sum_{t=0}^T \|\mathbf{y}_t - \mathbf{H}_t \mathbf{x}_t\|_{\mathbf{R}_t^{-1}}^2 \quad (1.11)$$

$$= \frac{1}{2} \|\mathbf{x}_0 - \mathbf{x}_b\|_{\mathbf{B}_0^{-1}}^2 + \frac{1}{2} \sum_{t=0}^T \|\mathbf{y}_t - \hat{\mathbf{h}}_t \mathbf{x}_0\|_{\mathbf{R}_t^{-1}}^2 \quad (1.12)$$

where the observations $\mathbf{y}_t$ and matrices $\mathbf{H}_t, \mathbf{R}_t$ are valid at time t , and $\hat{\mathbf{h}}_t = \mathbf{H}_t \mathcal{M}_{t-1} \mathcal{M}_{t-2} \dots \mathcal{M}_0$.

We will use the 4DEnVar method, where a control variable transform is used to write $\mathbf{x}_0$ as a linear combination of the ensemble perturbations:

$$\begin{aligned} \bar{\mathbf{x}}_b &= \frac{1}{N} \sum_{n=1}^N \mathbf{x}_b^{(n)} \\ \mathbf{X}'_b &= \frac{1}{\sqrt{N-1}} \left(\mathbf{x}_b^{(1)} - \bar{\mathbf{x}}_b, \mathbf{x}_b^{(2)} - \bar{\mathbf{x}}_b, \dots, \mathbf{x}_b^{(N)} - \bar{\mathbf{x}}_b \right) \\ \mathbf{x}_0 &= \bar{\mathbf{x}}_b + \mathbf{X}'_b \chi \end{aligned}$$

Here $\mathbf{x}_b^{(n)}$ represents the background state from ensemble member n , and N is the ensemble size. The cost function can then be rewritten in terms of χ and simplified by linearising $\hat{\mathbf{h}}_t$ around $\bar{\mathbf{x}}_b$:

$$J(\chi) = \frac{1}{2} \|\chi\|^2 + \frac{1}{2} \sum_{t=0}^T \left\| \mathbf{y}_t - \hat{\mathbf{h}}_t \bar{\mathbf{x}}_b - \hat{\mathbf{H}}_t \mathbf{X}'_b \chi \right\|_{\mathbf{R}_t^{-1}}^2 \quad (1.13)$$

where $\hat{\mathbf{H}}_t = \mathbf{H}_t \mathbf{M}_{t-1} \dots \mathbf{M}_0$ is the linearisation of $\hat{\mathbf{h}}_t$. Note that this cost function is further simplified since we have taken a background error covariance matrix of the form $\mathbf{B} = \mathbf{X}'_b \mathbf{X}'_b{}^\top$. This cost function has gradient

$$\nabla J(\chi) = \chi + \sum_{t=0}^T \left(\hat{\mathbf{H}}_t \mathbf{X}'_b \right)^\top \mathbf{R}_t^{-1} \left(\hat{\mathbf{h}}_t \mathbf{X}'_b \chi + \hat{\mathbf{H}}_t \bar{\mathbf{x}}_b - \mathbf{y}_t \right)$$

A major advantage of this formulation is that we can make the approximation

$$\hat{\mathbf{H}}_t \mathbf{X}'_b \approx \text{-----}$$

This eliminates the need to compute a tangent linear and adjoint model, which is a computationally expensive aspect of most variational data assimilation schemes ?.